8.2.4 Inhibitors of the Electron transport chain

ETC may be blocked by many inhibitors. ETC inhibitors act by binding a suitable area in the ETC. They prevent electrons from being passed from one carrier to the next one. Each inhibitor binds a particular carrier For example, rotenone and amytal inhibits complex I at NADH dehydrogenase and prevent NADH oxidation. Antimycin A and dimercaprol inhibit ETC at complex III. Poisons such as hydrogen sulphide, Cyanide, azide and carbon monoxide inhibit Complex IV. Oligomycin is a ATP synthase inhibitor. Many of the details of respiratory chain were obtained using inhibitors of ETC. The correct order of the ETC components was determined using inhibitors. Using oxygen electrode, the extent of electron transport was measured. During this process, when electron transport is inhibited, oxygen consumption is diminished. (Fig.8.9).

8.3 OXIDATIVE PHOSPHORYLATION

During aerobic cellular respiration, most of the ATP is generated by oxidative phosphorylation where electrons derived from NADH and $FADH_2$ are oxidised by electron transport through the respiratory chain. In oxidative phosphorylation, carbon fuels are oxidized in TCA cycle to give electrons with high transfer potential. Eventually, this electro-motive force is converted into a proton-motive force and finally into phosphoryl transfer potential.

The conversion of electro-motive force into proton-motive force is carried out by three electron-driven proton pumps: NADH-UQ oxidoreductase, UQ-cytochrome c oxidoreductase, and cytochrome c oxidase. These large transmembrane complexes contain multiple oxidation-reduction centers, including quinones, flavins, iron-sulfur clusters, hemes, and copper ions. The final phase of oxidative phosphorylation is carried out by ATP synthase, an ATP-synthesizing assembly that is driven by the flow of protons back into the mitochondrial matrix. Components of this remarkable enzyme rotate as part of its catalytic mechanism.

Oxidative phosphorylation clearly shows that proton gradients are an interconvertible currency of free energy in biological systems. Oxidative phosphorylation is different from substrtate level phosphorylation, as it does not involve phosphorylated chemical intermediates and uses a different mechanism known as chemiosmotic hypothesis.

8.3.1 Chemiosmotic theory

In 1961, Peter Mitchell proposed the Chemiosmotic theory which demonstrates that the energy derived from oxidation of components in the ETC is coupled to the translocation of protons from the inside to the outside of the inner surface of the mitochondrial membrane. The electrochemical potential difference resulting from the asymmetric distribution of the hydrogen ions is used to drive the mechanism responsible for the formation of ATP. As explained earlier, the complexes of the respiratory chain act as proton pumps.

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